

# PRATEET TIWARI

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## EXPERIENCE

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### Frontend Engineer Intern | [TechSnap](#)

Mar 2026 – May 2026

Bengaluru, India (Remote)

- Developed high-performance dashboards using **Next.js**, optimizing core web vitals and overall SEO metrics.
- Optimized application server-state synchronization with **TanStack Query**, reducing redundant API network requests by caching queries.
- Implemented structured client-side state slices via **Redux Toolkit**, maximizing UI rendering predictability across complex user dashboards.

## PROJECTS

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### [Video Streaming Platform](#) | *Go, Gin, Asynq, AWS S3, Redis Streams, FFmpeg, Next.js, TanStack Query*

- Built a distributed video processing pipeline using **Go**, **Asynq**, and **FFmpeg** workers for asynchronous transcoding and media processing.
- Implemented adaptive bitrate **HLS** streaming with parallel transcoding using **Goroutines** and **Semaphore**-based concurrency control for efficient multi-resolution video processing.
- Architected a distributed view-counting pipeline using **Redis Streams**, **Consumer Groups**, heartbeat validation, **Redis HyperLogLog**, deduplication, and **cron**-driven batched hot-counter synchronization to **MongoDB**.

### [Distributed Notification System](#) | *TypeScript, NestJS, BullMQ, Redis, Nginx, Firebase, Twilio, Resend*

- Built a distributed notification platform supporting **Email**, **SMS**, **Push**, and **In-App** delivery channels.
- Designed an asynchronous delivery pipeline using **BullMQ** and **Redis**, implementing retries, delivery tracking, user preferences, and worker-based processing.
- Implemented real-time in-app notifications using **SSE** and **Redis Pub/Sub**, integrated **Firebase FCM**, **Twilio**, and **Resend**, and added **Nginx** load balancing with CI automation via **GitHub Actions**.

### [Secure Payment Microservice](#) | *Node.js, PostgreSQL, Prisma, Stripe, BullMQ, Redis*

- Built a **race-condition safe** payment system using **Stripe** webhooks as the sole source of truth.
- Handled 50+ concurrent requests with zero duplicate charges using **idempotent APIs** and database constraints.
- Validated reliability through concurrent load testing, ensuring idempotent request handling, retry mechanisms, and **Dead-letter Queue (DLQ)** recovery.

## OPEN SOURCE

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[Cal.com](#) | Fixed App Store search inconsistencies through input normalization ([PR #29236](#))

[Directus](#) | Fixed incorrect pie chart tooltip values, improving analytics and data accuracy ([PR #27356](#))

## TECHNICAL SKILLS

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- **Languages:** TypeScript, Go, JavaScript
- **Backend:** Node.js, Express, NestJS, Gin, Socket.io, WebRTC, BullMQ, Asynq
- **Frontend:** React.js, Next.js, TanStack Query, Redux Toolkit, Tailwind CSS
- **Databases & Storage:** MongoDB, PostgreSQL, Redis, ClickHouse, AWS S3
- **Cloud & DevOps:** Docker, Git, GitHub Actions, Nginx, Vercel, Render

## EDUCATION

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**ABES Engineering College**, Ghaziabad, India

Expected Aug 2027

Bachelor of Technology in Computer Science and Engineering (CSE)